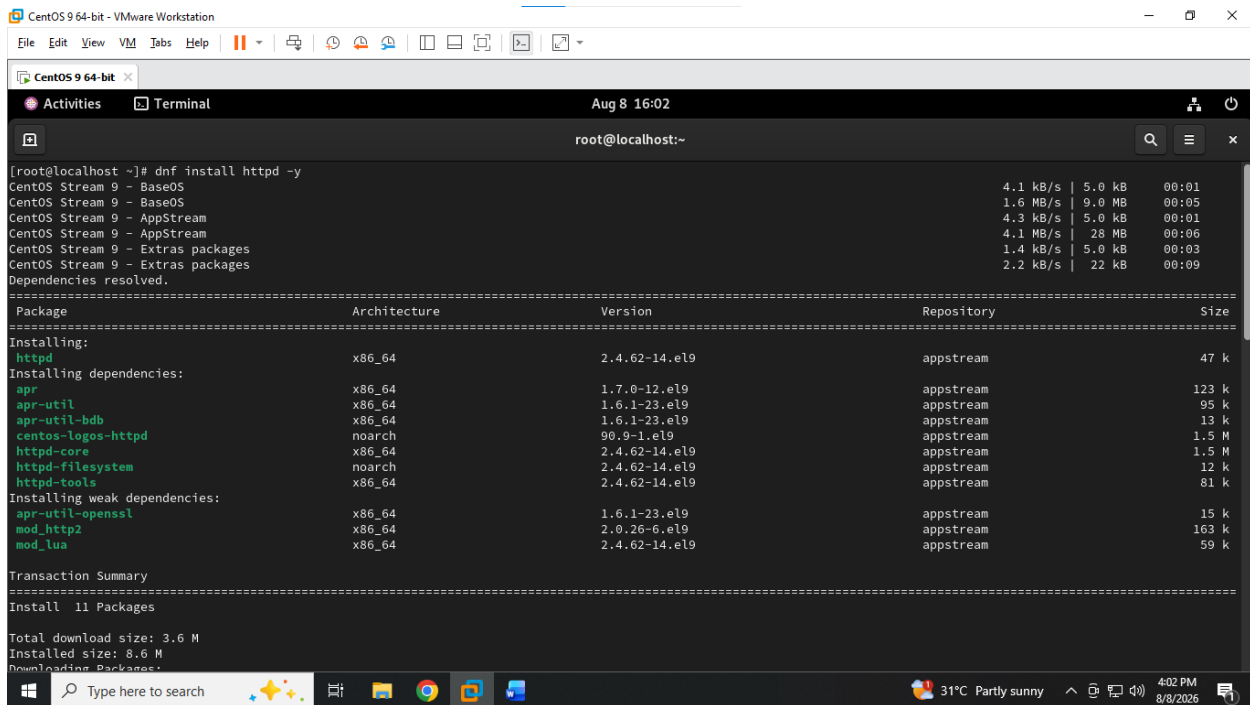


Module-1 : Introduction to Linux Services

Guided by : Vipul mistri

Q.1 Install the Apache web server on your Linux machine using the package manager (apt or yum), and confirm the installation by running the command to check its version.

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 8 16:02
root@localhost:~

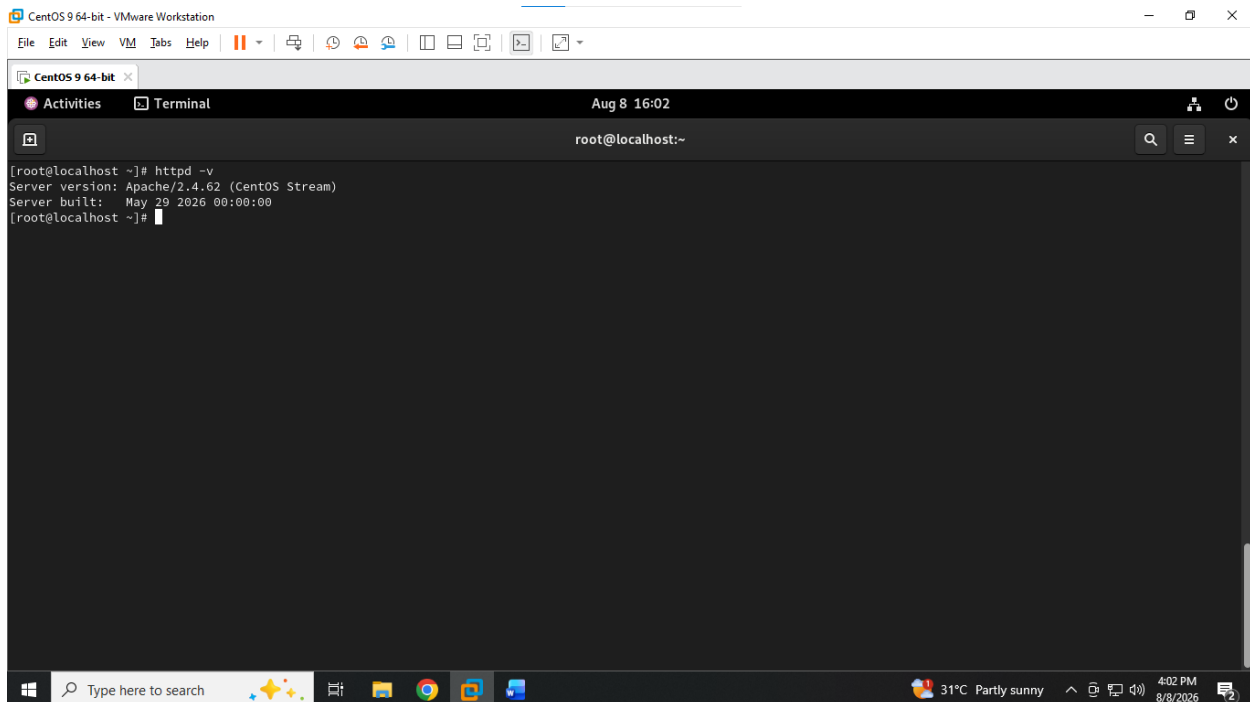
[root@localhost ~]# dnf install httpd -y
CentOS Stream 9 - BaseOS                               4.1 kB/s | 5.0 kB   00:01
CentOS Stream 9 - BaseOS                               1.6 MB/s | 9.0 MB   00:05
CentOS Stream 9 - AppStream                             4.3 kB/s | 5.0 kB   00:01
CentOS Stream 9 - AppStream                             4.1 MB/s | 28 MB    00:06
CentOS Stream 9 - Extras packages                       1.4 kB/s | 5.0 kB   00:03
CentOS Stream 9 - Extras packages                       2.2 kB/s | 22 kB    00:09
Dependencies resolved.

=====
Package                        Architecture      Version           Size
=====
Installing:
httpd                          x86_64            2.4.62-14.el9     47 k
Installing dependencies:
apr                            x86_64            1.7.0-12.el9      123 k
apr-util                      x86_64            1.6.1-23.el9       95 k
apr-util-bdb                  x86_64            1.6.1-23.el9       13 k
centos-logos-httpd            noarch            90.9-1.el9         1.5 M
httpd-core                    x86_64            2.4.62-14.el9     1.5 M
httpd-filesystem              noarch            2.4.62-14.el9      12 k
httpd-tools                   x86_64            2.4.62-14.el9       81 k
Installing weak dependencies:
apr-util-openssl              x86_64            1.6.1-23.el9       15 k
mod_http2                     x86_64            2.0.26-6.el9       163 k
mod_lua                       x86_64            2.4.62-14.el9        59 k

Transaction Summary
=====
Install 11 Packages

Total download size: 3.6 M
Installed size: 8.6 M
Downloaded Packages:

=====
Type here to search 31°C Partly sunny 4:02 PM 8/8/2026
```



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 8 16:02
root@localhost:~

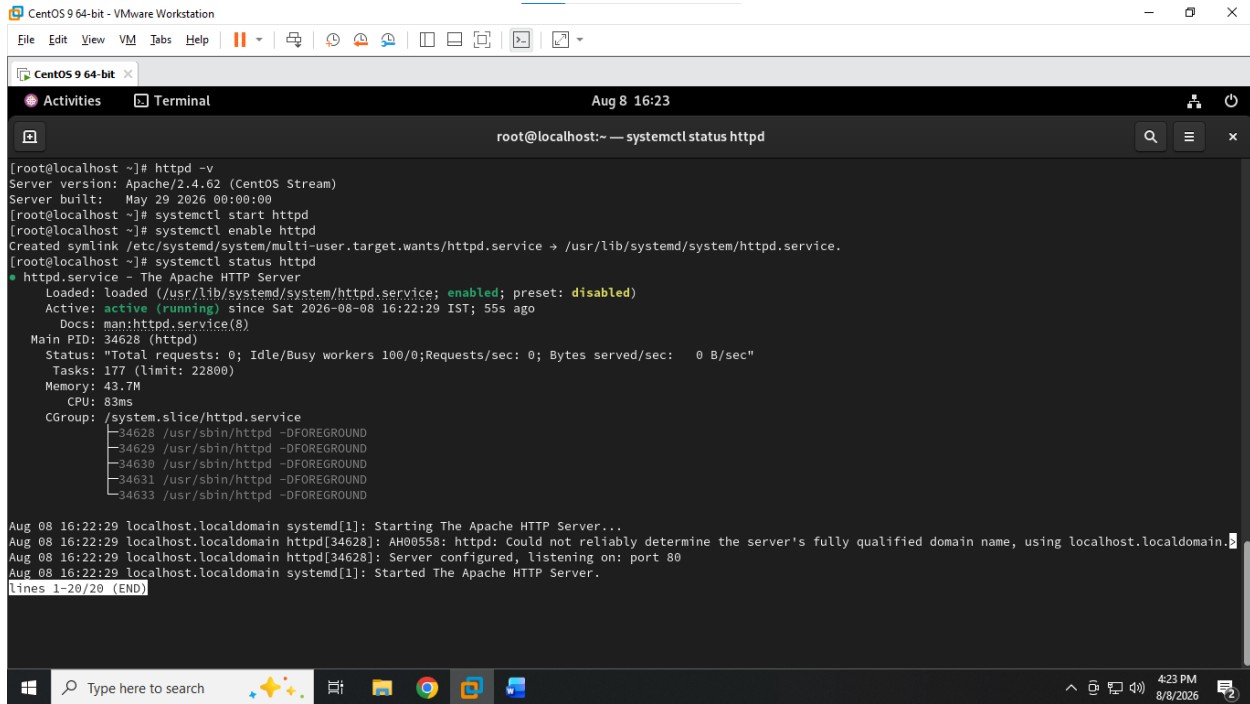
[root@localhost ~]# httpd -v
Server version: Apache/2.4.62 (CentOS Stream)
Server built: May 29 2026 00:00:00
[root@localhost ~]#
```

Module-1 : Introduction to Linux Services

Guided by : Vipul mistri

Q.2 Start the Apache service using systemctl, then use systemctl status to verify it is running

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 8 16:23
root@localhost:~ -- systemctl status httpd

[root@localhost ~]# httpd -v
Server version: Apache/2.4.62 (CentOS Stream)
Server built: May 29 2026 00:00:00
[root@localhost ~]# systemctl start httpd
[root@localhost ~]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.
[root@localhost ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-08-08 16:22:29 IST; 55s ago
     Docs: man:httpd.service(8)
   Main PID: 34628 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 22800)
   Memory: 43.7M
     CPU: 83ms
   CGroup: /system.slice/httpd.service
           └─34628 /usr/sbin/httpd -DFOREGROUND
             └─34629 /usr/sbin/httpd -DFOREGROUND
               └─34630 /usr/sbin/httpd -DFOREGROUND
                 └─34631 /usr/sbin/httpd -DFOREGROUND
                   └─34633 /usr/sbin/httpd -DFOREGROUND

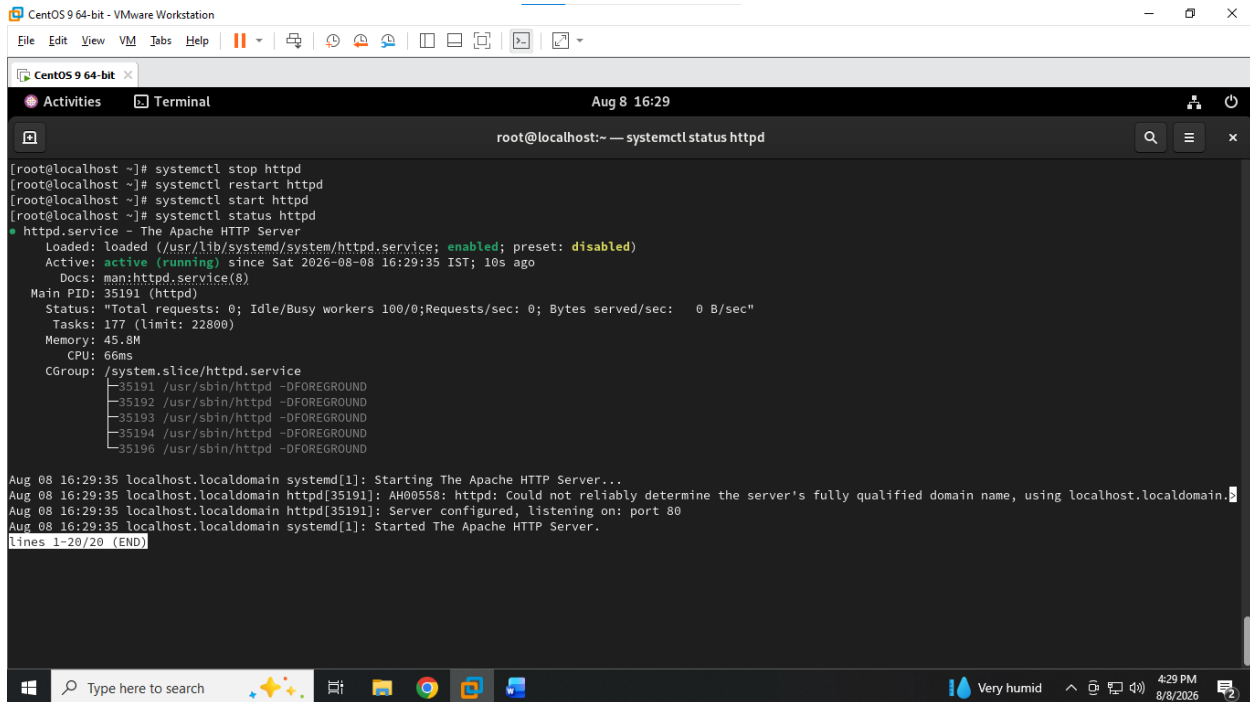
Aug 08 16:22:29 localhost.localdomain systemd[1]: Starting The Apache HTTP Server...
Aug 08 16:22:29 localhost.localdomain httpd[34628]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using localhost.localdomain.
Aug 08 16:22:29 localhost.localdomain httpd[34628]: Server configured, listening on: port 80
Aug 08 16:22:29 localhost.localdomain systemd[1]: Started The Apache HTTP Server.
lines 1-20/20 (END)
```

Module-1 : Introduction to Linux Services

Guided by : Vipul mistri

Q.3 Stop and restart the Apache service using systemctl, and observe the difference between stop, start, and restart commands. Hint Use 'systemctl stop apache2', 'systemctl start apache2', and 'systemctl restart apache2' (or 'httpd' on some systems).

ANS :



```
CentOS 9 64-bit - VMware Workstation
File Edit View VM Tabs Help
CentOS 9 64-bit x
Activities Terminal Aug 8 16:29
root@localhost:~ — systemctl status httpd

[root@localhost ~]# systemctl stop httpd
[root@localhost ~]# systemctl restart httpd
[root@localhost ~]# systemctl start httpd
[root@localhost ~]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-08-08 16:29:35 IST; 10s ago
     Docs: man:httpd.service(8)
   Main PID: 35191 (httpd)
   Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 22800)
   Memory: 45.8M
     CPU: 66ms
   CGroup: /system.slice/httpd.service
           └─35191 /usr/sbin/httpd -DFOREGROUND
             └─35192 /usr/sbin/httpd -DFOREGROUND
               └─35193 /usr/sbin/httpd -DFOREGROUND
                 └─35194 /usr/sbin/httpd -DFOREGROUND
                   └─35196 /usr/sbin/httpd -DFOREGROUND

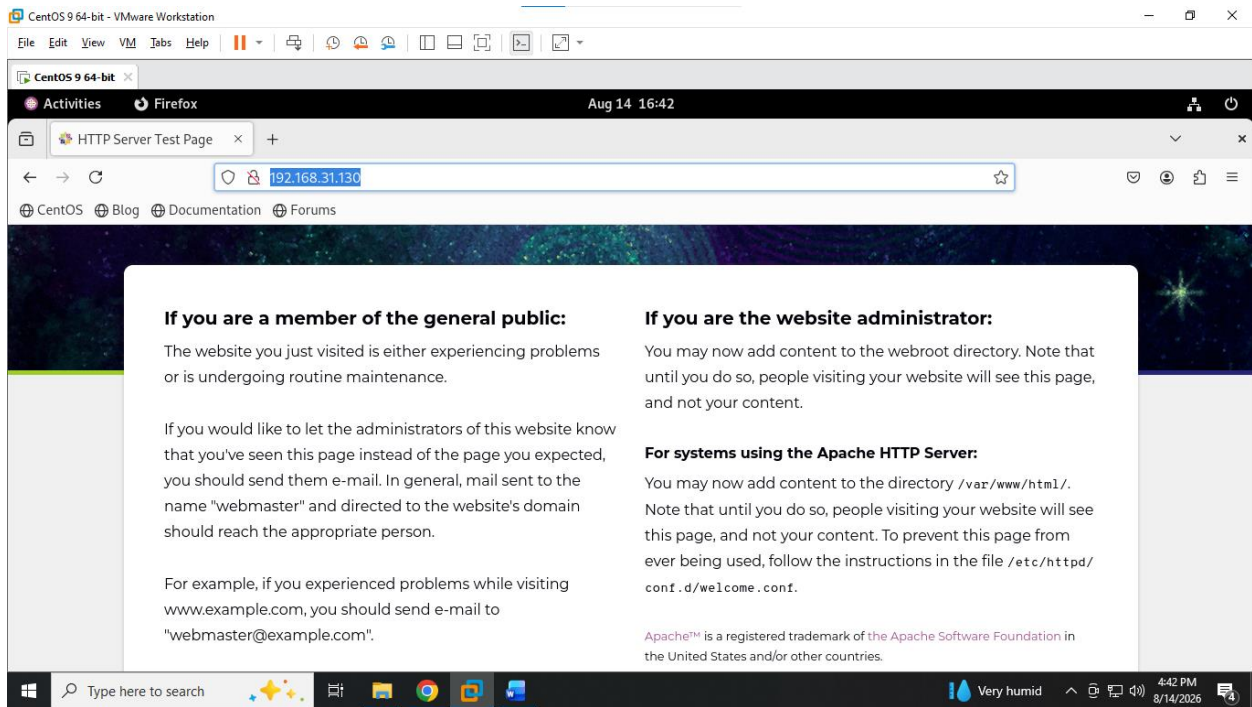
Aug 08 16:29:35 localhost.localdomain systemd[1]: Starting The Apache HTTP Server...
Aug 08 16:29:35 localhost.localdomain httpd[35191]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using localhost.localdomain.
Aug 08 16:29:35 localhost.localdomain httpd[35191]: Server configured, listening on: port 80
Aug 08 16:29:35 localhost.localdomain systemd[1]: Started The Apache HTTP Server.
lines 1-20/20 (END)
```

Module-1 : Introduction to Linux Services

Guided by : Vipul mistri

4. Open your browser and access <http://localhost> or your Linux machine's IP address to verify that the Apache default web page is displayed.

ANS :



Module-1 : Introduction to Linux Services

Guided by : Vipul mistri

Q.5 Write a short comparison (3-4 lines) between Apache and Nginx web servers, mentioning one key use-case for each based on your research.

ANS : Apache and Nginx are both popular web servers, but they differ in architecture and typical use cases.

Apache is highly flexible and is commonly used for dynamic websites and applications requiring .htaccess support.

Nginx is lightweight and efficient, making it well suited for serving static content and acting as a reverse proxy or load balancer.

In short, Apache emphasizes flexibility, while Nginx emphasizes performance and efficient handling of many concurrent connections.